

Principal component Analysis

Key points:

$$\text{Var}(PC_1) > \text{Var}(PC_2) > \text{Var}(PC_3) \dots$$

Number of PC depends on vector dimension

Say you want to convert 3D vector to 2D

find $PC_1, PC_2 \dots PC_k$

find Projection to PC_1 & PC_2

Those are your new 2D vectors

Using (PCs) (3D $\rightarrow$ 2D)

$$\text{Say } PC_1 = \begin{bmatrix} 0.707 \\ 0.707 \\ 0 \end{bmatrix}$$

$$PC_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$PC_3 = \begin{bmatrix} 0.707 \\ -0.707 \\ 0 \end{bmatrix}$$

$$\text{Say data point} = \begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix}$$

find co-ordinate along PC_1

$$X \cdot PC_1 =$$

$$4 \times (0.707) + 2 \times (0.707) + 3 \times (0) = 4.242$$

$$PC_2$$

$$= X \cdot PC_2 = 3$$

$$PC_3 = X \cdot PC_3 = 1.414$$

$$\text{New vector} = (4.242, 3, 1.414)$$

Say you want 3D to 2D $\rightarrow (4.242, 3)$

PC change coordinates.

Check how much variance was capture

Eigen value	PC
PC1	8
PC2	1.5
PC3	0.5

$$PC_{total} = 8 + 1.5 + 0.5 = 10$$

$$PC_1 \text{ captures} = \frac{8}{10} \times 100 = 80 \text{ percent}$$

$$PC_2 \text{ captures} = \frac{1.5}{10} \times 100 = 15\%$$

$$PC_3 \text{ capture} = \frac{0.5}{10} \times 100 = 5\%$$